

Phoenix Cyber Resilience Project

CY020 – Data Protection Framework

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1. INTRODUCTION

The Phoenix Cyber Resilience Project handles different forms of operational, security, and governance-related information that require strong protection measures. As cyber threats continue to evolve, organisations must implement effective data protection mechanisms to ensure confidentiality, integrity, and availability of information assets.

This document presents the proposed Data Protection Framework developed for the Phoenix project under the Governance & Compliance stream. The framework outlines the processes, policies, and security controls required to protect sensitive data from unauthorised access, misuse, loss, or cyber attacks. The document also aligns with recognised cyber security standards and industry best practices to support secure system operations and long-term cyber resilience.

2. PURPOSE OF DATA PROTECTION

Data protection is critical for maintaining trust, ensuring business continuity, and preventing security incidents that may compromise sensitive information. The Phoenix project relies on secure handling of system data, threat intelligence information, user records, documentation assets, and monitoring logs.

The primary objectives of this framework include:

- Protecting confidential information from unauthorised access
- Ensuring integrity and accuracy of stored data
- Supporting secure communication and collaboration
- Reducing risks related to cyber attacks and data breaches
- Supporting compliance with cybersecurity standards and privacy regulations
- Improving organisational resilience during incidents or system failures

Strong data protection practices also support governance activities by ensuring accountability, traceability, and secure management of project documentation and digital assets.

3. TYPES OF DATA WITHIN PHOENIX

The Phoenix system may process and store multiple categories of information. Different levels of security controls are required depending on the sensitivity and criticality of the data.

Data Type	Description	Sensitivity Level
User Credentials	Login usernames and passwords	High
Threat Intelligence Data	Threat feeds and indicators	High
System Logs	Security and monitoring logs	Medium
Governance Documents	Policies, procedures, compliance files	Medium
Employee Information	Team member records and details	High
Backup Files	Recovery and archived data	High
Public Documentation	Non-sensitive public information	Low

Proper identification and categorisation of data allows the project team to apply appropriate protection measures based on risk exposure and operational importance.

4. DATA CLASSIFICATION FRAMEWORK

To improve security governance, all information assets within the Phoenix project should follow a data classification model. This ensures consistent handling, storage, and access management practices.

Classification Levels

Classification	Description	Example
Public	Information approved for public access	General project overview
Internal	Information limited to project members	Internal meeting notes
Confidential	Sensitive operational information	Security reports
Restricted	Highly sensitive critical information	Credentials and encryption keys

Restricted and Confidential data require stronger security controls including encryption, limited access permissions, and continuous monitoring.

5. ACCESS CONTROL & AUTHENTICATION

Access control mechanisms play an essential role in protecting sensitive information from unauthorised access. The Phoenix project should implement Role-Based Access Control (RBAC) to ensure users only access information necessary for their responsibilities.

Recommended Security Controls

- Role-Based Access Control (RBAC)
- Multi-Factor Authentication (MFA)
- Strong password enforcement
- Principle of Least Privilege
- Account monitoring and auditing
- Session timeout mechanisms

The Principle of Least Privilege ensures that users are granted only the minimum permissions required to complete authorised tasks. This reduces the risk of insider threats, accidental exposure, and privilege abuse.

6. DATA ENCRYPTION & SECURE STORAGE

Encryption is necessary to protect sensitive information during storage and transmission. The Phoenix project should implement modern encryption techniques to secure data across all systems and communication channels.

Proposed Encryption Controls

Security Area	Recommended Method
Data at Rest	AES-256 Encryption
Data in Transit	TLS/HTTPS
Password Storage	Secure hashing with salting
Cloud Storage	Encrypted cloud repositories

Encryption significantly reduces the likelihood of attackers accessing readable information even if a system compromise occurs.

Secure storage environments should also include:

- Access logging
- Backup redundancy
- Secure cloud configuration
- File integrity monitoring

7. BACKUP & DISASTER RECOVERY

Data backups are important for maintaining operational continuity during cyber incidents, hardware failures, or accidental data loss.

Recommended Backup Strategy

- Daily automated backups
- Weekly full-system backups
- Cloud and offline backup redundancy
- Periodic backup testing
- Secure backup encryption

The project should also maintain a Disaster Recovery Plan (DRP) outlining recovery procedures, recovery priorities, and restoration timelines following a major system disruption.

8. DATA RETENTION & DISPOSAL

Information should only be retained for as long as operationally necessary. Improper storage of outdated data may increase security risks and compliance concerns.

Recommended Practices

- Define retention periods for each data category
- Securely archive inactive information
- Permanently destroy obsolete sensitive data
- Apply secure wiping techniques for storage devices
- Maintain disposal records for audit purposes

Secure disposal reduces the risk of data leakage and unauthorised recovery of sensitive information.

9. COMPLIANCE & REGULATORY ALIGNMENT

The Phoenix Data Protection Framework aligns with recognised cybersecurity and privacy standards to improve governance and operational security.

Relevant Standards & Frameworks

Standard / Framework	Purpose
ISO/IEC 27001	Information Security Management
NIST Cybersecurity Framework	Risk management and security guidance
Australian Privacy Principles (APP)	Privacy protection requirements
Essential Eight	Baseline cyber security mitigation strategies

Compliance with recognised frameworks improves organisational credibility and strengthens overall cyber resilience.

10. RISK MANAGEMENT CONSIDERATIONS

Several cyber security risks may impact the confidentiality, integrity, and availability of project data.

Key Risks

Risk	Impact
Phishing Attacks	Credential theft
Ransomware	Data encryption and downtime
Insider Threats	Unauthorised information disclosure
Weak Passwords	Account compromise
Misconfigured Storage	Data exposure

Risk mitigation strategies include continuous monitoring, security awareness training, vulnerability management, and regular compliance reviews.

11. RECOMMENDATIONS

The following recommendations are proposed to strengthen data protection within the Phoenix project:

- Enforce MFA across all systems
- Conduct regular security audits
- Implement automated backup monitoring
- Perform periodic access reviews
- Introduce staff cyber security awareness training
- Apply encryption to all sensitive data assets
- Develop formal incident response procedures

These recommendations will support improved governance maturity and reduce exposure to cyber threats.

12. CONCLUSION

The Data Protection Framework developed for the Phoenix Cyber Resilience Project provides a structured approach to securing sensitive information and supporting effective governance practices. Through strong access controls, encryption mechanisms, backup strategies, and compliance alignment, the framework helps improve confidentiality, integrity, and availability of organisational data assets.

Implementing these controls will strengthen cyber resilience, minimise operational risks, and support secure collaboration within the Phoenix project environment. Continuous improvement and regular review of security practices will remain essential as cyber threats continue to evolve.